

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2024

CS1231-1 – INTRODUCTION TO DATA SCIENCE

(For CS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. In _____, data from the entire population or a sample is summarized with numerical descriptors.

A) Inferential statistics	B) Text mining
C) Descriptive statistics	D) Sentimental analysis
2. What is the primary purpose of Business Intelligence (BI)?

A) To clean and preprocess data	B) To present historical data to aid decision-making
C) To perform data mining	D) To transform data
3. What is data mining primarily concerned with?

A) Collecting raw data	B) Cleaning and processing data
C) Identifying patterns and relationships in data	D) Reporting and visualization
4. What is the first step in the data analysis process?

A) Data Collection	B) Data Cleaning
C) Data Conversion	D) Data Requirements Specification
5. The _____ function in excel calculates the last day of a month by adding a specified number of months to a given date.

A) DAY	B) DAYS
C) EOMONTH	D) WEEKS
6. Normal formatting is _____, while conditional formatting is _____.

A) static and static	B) dynamic and dynamic
C) static and dynamic	D) dynamic and static
7. The formula = REPLACE("Python Excel",1,6,"MS") produces output as?

A) MS	B) MS Excel
C) Excel	D) Python MS
8. What does 'Filtering' data do in excel?

A) Changes the data formatting	B) Deletes unwanted data
C) Temporarily hides rows or columns that do not meet the specified criteria	D) Permanently removes data that doesn't meet a criterion
9. What does the VLOOKUP stands for

A) Visual Lookup	B) Vertical Lookup
C) Horizontal Lookup	D) Virtual Lookup
10. Which chart type is best for showing trends over time?

A) Pie chart	B) Line chart
C) Bar chart	D) Scatter plot

11. Which feature allows you to drill down into the details of a PivotTable summary?
- A) Data Validation B) Expand/Collapse
C) Scenario Manager D) Solver
12. Which feature in PivotTable helps in showing the percentage of total values?
- A) Data Bars B) Value Field Settings
C) Quick Analysis D) Conditional Formatting
13. Which of the following is used to install a package in R?
- A) download.package() B) install.packages()
C) install() D) get()
14. Assume $b \leftarrow 2:7$. What is the length of b ?
- A) 0 B) 7
C) 5 D) 6
15. In R language when the vectors have different lengths, the multiplication begins with the _____ vector and continues till all the elements in the _____ vector have been multiplied.
- A) smaller and smaller B) smaller and larger
C) larger and smaller D) larger and larger
16. Assume R-object containing $a = \text{list}(1, "Orange", \text{list}(2, "Apple"))$. How to access "Apple" from list a ?
- A) $a[[3]][2]$ B) $a[2]$
C) $a[3]$ D) $a[3:2]$
17. Which method in R is used to create multiple line graphs?
- A) box() B) plots()
C) lines() D) double()
18. A _____ depicts information like the minimum and maximum data point, the median value, first and third quartile, and interquartile range.
- A) Boxplot B) Bar plot
C) Scatter plot D) Pie plot
19. Which method is used to draw only the points while plotting a line graph?
- A) Plot (data) B) Plot(data, type = "o")
C) Plot (data, type = "l") D) Plot (data, type = "p")
20. _____ depicts whether an association exists between bivariate data.
- A) Boxplot B) Scatter plot
C) Bar plot D) Pie plot

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	With a small example discuss any of the <i>four</i> text functions used in Excel:	6	L*2	2	1
b)	Define the following terms i. Data Mining ii. Business Intelligence iii. Predictive Analytics iv. Text Analytics	4	L1	1	1
c)	Given the following scenario, use Excel date functions to determine the project end dates and explain each step: Project Start Date: 1st January 2024 (in cell A1) Total Work Days Assigned: 15 days Weekend Days: Friday and Saturday Public Holidays: 26th January 2024 (in cell B1). Write functions to perform the following tasks: (i) Calculate the project end date assuming standard weekends (Saturday and Sunday). (ii) Calculate the project end date with weekends set as Friday and Saturday.	6	L3	2	2

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|-------|--|---|----|---|---|
| 2. a) | List and explain any <i>four</i> categories available in Excel's Conditional Formatting, and provide an example for each category. | 8 | L2 | 2 | 1 |
| b) | Given a dataset of employees with columns for Employee ID, Name, Department, Salary, and Date of Joining describe the steps needed to perform each of the following tasks using Excel: | | | | |
| i. | Sort the dataset in ascending order of Salary and then by Date of Joining within each salary group. | | | | |
| ii. | Filter the dataset to show only employees from the IT department with a salary greater than 60,000. | 4 | L2 | 2 | 2 |
| c) | Using Excel functions, perform the following tasks with a sales dataset that includes fields for Salesperson, Region, and Amount: | | | | |
| i. | Determine the number of sales made in the "North" region. | | | | |
| ii. | Calculate the total sales amount made by the salesperson named "John." | 4 | L3 | 2 | 2 |
| 3. a) | Describe the process of using the Subtotal feature in Excel to summarize data. Include a step-by-step explanation with an example. | 4 | L2 | 2 | 1 |
| b) | List and explain at least <i>four</i> Time functions in Excel with example for each. | 4 | L2 | 2 | 1 |
| c) | Given a dataset containing columns for Song Number, Singer, Date, and Minutes, write Excel formulas to compute the following: | | | | |
| i. | The number of songs that are at least 4 minutes long. | | | | |
| ii. | The number of songs that are longer than the average length of all songs in the list. | | | | |
| iii. | The number of songs sung by singers whose names begin with "S." | | | | |
| iv. | The number of songs sung by singers whose names contain exactly 6 characters. | | | | |
| v. | The number of songs that were played after June 15, 2005. | | | | |
| vi. | The number of songs sung by Bruce Springsteen that are exactly 4 minutes long. | | | | |
| | | 8 | L3 | 2 | 2 |

Unit – II

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|-------|---|---|----|---|---|
| 4. a) | Explain two lookup functions in Excel and demonstrate how to use them to retrieve data from an employee dataset containing the fields EmployeeID, Department, and Designation. | 6 | L2 | 2 | 1 |
| b) | Explain the process of Data Consolidation in Excel and its advantages. | 6 | L2 | 2 | 1 |
| c) | Explain the process of creating a Data Model in Excel and establishing relationships between different datasets. | 4 | L2 | 3 | 1 |
| 5. a) | What is Pivot Table in excel? Assume you have a dataset with Salesperson, Region, Product, and Sales Amount . Explain the application and steps for using Pivot Table on the same dataset. | 6 | L2 | 3 | 1 |
| b) | Discuss any <i>three</i> common type of Data visualization techniques used in excel. | 6 | L2 | 3 | 1 |
| c) | Write a note on Exploring data with Power Pivot. | 4 | L2 | 3 | 1 |
| 6. a) | Describe any <i>four</i> functions used for financial analysis in excel. | 8 | L2 | 3 | 1 |
| b) | What are KPIs? Identify the possible KPIs with an example of a student dataset which includes fields such as StudentID, Name, Grades, and Attendance. | 8 | L3 | 3 | 2 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | Define the following in R with a proper syntax and examples | | | | |
| | i. Array ii. Matrix | | | | |
| | ii. Vector iv. Factors | 8 | L2 | 4 | 1 |
| b) | Describe the different methods for importing data into R from external files, such as CSV, JSON, XML, and Web | 8 | L2 | 4 | 1 |
| 8. a) | Explain different types of argument matching for function call in R with example or each. | 8 | L2 | 4 | 1 |
| b) | Discuss the following in R | | | | |
| | i. Scatterplot | | | | |
| | ii. Histograms | | | | |
| | iii. Boxplot. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
